

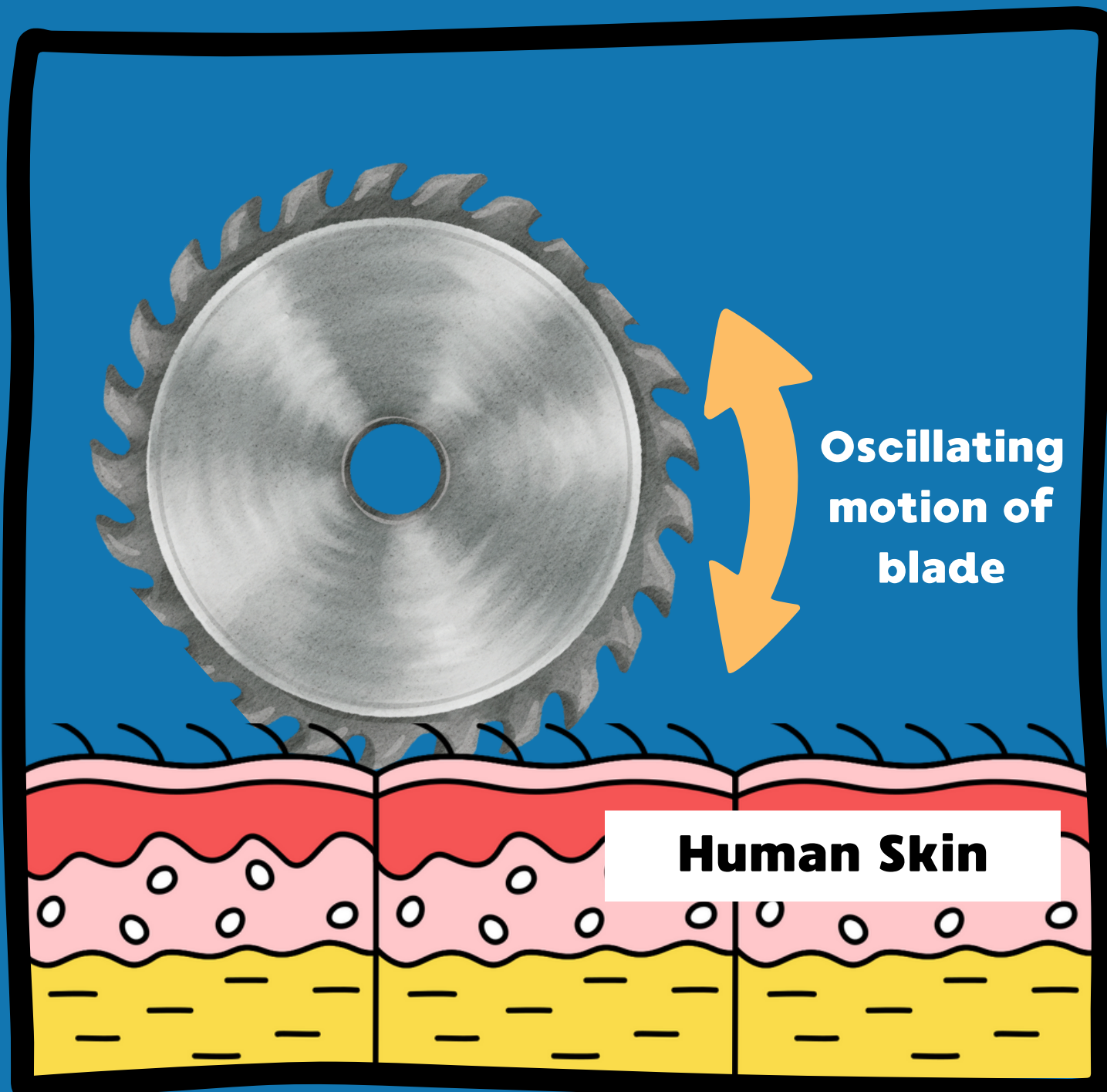
A CUT ABOVE THE REST: THE SCIENCE BEHIND CAST SAWS

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01

Why won't I get cut?

- Unlike regular saws which have rotating blades, cast saws use a small-toothed blade that **oscillates** very rapidly over a **tiny angle**.
- This oscillating motion easily **cuts rigid materials** (e.g., fiber glass, plaster) by inducing **microfractures** that ultimately cause failure & fracture.
- Human skin is **soft and elastic**, so it **moves along** with the vibrations of the blade, meaning it won't get cut.



03

Risks & Precautions

Cast saws are generally safe for the skin but are not completely risk-free. Improper use, such as applying too much pressure or keeping the blade in one place for too long **increases friction and heat**. Excess heat and friction could potentially result in **skin burns**.



02

Features of a cast saw

- The oscillation is generated by **vibration motors**
- The blade typically oscillates between **180-400 Hz**
- The cast saw also comes with other handy features such as a **vacuum system** to efficiently remove dust and debris

